Knowledge about typical events, and not morphological cues, facilitates the processing of English adjective-noun constructions

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We examine how world knowledge and linguistic knowledge (morphological and semantic cues) facilitate the retrieval of semantic information associated with nouns in on-line processing, using adjective-noun constructions such as *skillful dancer* as a case study. We find that eventive nouns facilitate retrieval of the skill in question in these constructions, with no advantage for deverbal over nondeverbal nouns, highlighting the contribution of semantic over morphological information in the identification of events.

Background: Adjectives like *skillful* and *talented* are underspecified regarding the skill in question. When combining a *skillful*-type adjective with an individual noun, default readings may or may not arise, depending on the noun type. These adjectives can thus serve as a 'window' into noun meaning: while *skillful dancer*/*ballerina* (deverbal/nondeverbal eventives) can be paraphrased as *skillful as a dancer*/*ballerina*, *skillful woman* (a personal stative) cannot be paraphrased as *skillful as a woman*. When computing meaning for the latter, comprehenders cannot fall back on the default interpretation supplied by the noun but have to rely on world knowledge. In semantics, the source of the default reading is captured by the presence of a nominal event argument in deverbal nouns which associates an event (e.g. *dancing*) with a noun's meaning. Event arguments are often not assumed in nondeverbal eventives such as *ballerina* [1,2], while other proposals take them as also semantically associated with events [3]. Personal statives are not assumed to have an event argument. The two approaches make differing predictions regarding the ease of skill retrieval when linking this to RTs in behavioral experiments. If event arguments facilitate retrieval, deverbal and nondeverbal eventives will be processed faster than personal statives. If deverbality alone facilitates retrieval, deverbal eventives will be processed faster than both nondeverbal eventives and personal statives. If the two factors (deverbality and the presence of an event argument) interact, deverbal eventives will be processed faster than nondeverbal eventives, which will be processed faster than personal statives.

Our maze **experiment** [4] ($N_{subject}$ = 70, N_{item} = 24, N_{filler} = 48) tested these predictions using English stimuli such as in (1), following a 3x1 design manipulating the factor *Noun type* with the levels *Deverbal eventive*, *Nondeverbal eventive*, and *Personal stative*. All nouns used were previously normed in a speeded acceptability judgment task, ensuring the nouns in fact correspond to their respective assigned class. The second sentence in each maze task stimulus targeted the skill/talent introduced in the first sentence using a *good at it*-construction. *It* targets the skill in the previously introduced adjective-noun construction. We hypothesize that the ease with which the skill can be accessed to anaphorically resolve the pronoun *it* is linked to the difficulty that comprehenders have in retrieving the skill in the adjective-noun combination.

Results: Figure 1 shows average word-by-word RTs in the maze task. We fit a Bayesian linear mixed model to the log-transformed RTs in the critical region, *it*. Helmert-coded contrasts showed no difference between deverbal and nondeverbal eventives, whereas their grand mean differed from the personal statives (see Table 1), with this effect persisting even past the critical region.

Discussion: Our results suggest that event arguments, a semantic cue, facilitate skill retrieval, with no advantage for deverbal nouns, even though both speed up skill retrieval compared to computations involving personal statives and relying on world knowledge only. While our results are compatible with a semantic account based on event arguments, an alternative explanation could be that the world knowledge associated with different individual nouns differs, with more specific world knowledge associated with eventive nouns facilitating skill retrieval over skill

retrieval for personal statives, for which less specific knowledge regarding possible activities is available.

(1) Sample stimuli from our experiment:
   a) *Riley is a skillful **dancer**. She is really good at it before large crowds* (deverbal eventive).
   b) *Riley is a skillful **ballerina**. She is really good at it before large crowds* (nondeverbal eventive).
   c) *Riley is a skillful **person**. She is really good at it before large crowds* (personal stative).

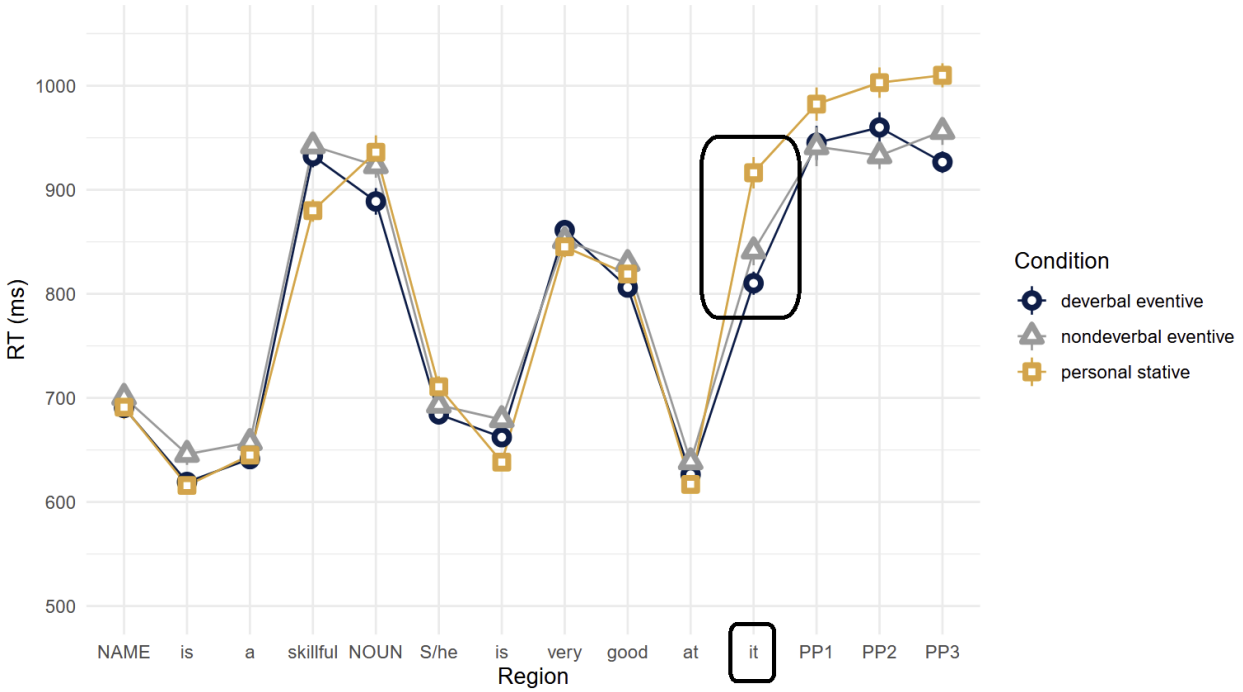

Figure 1: Word-by-word RTs. *NOUN* represents the noun class manipulated; *it* is the critical region. Error bars represent Cousineau-corrected SEs.

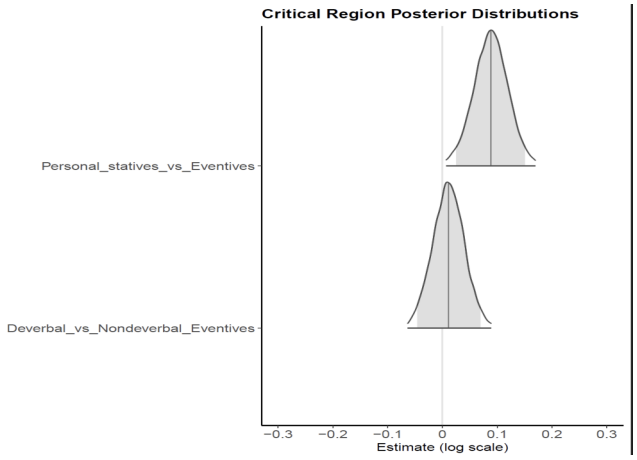

Figure 2: Posterior distributions over RTs in the critical region, for the contrast between Deverbal and Nondeverbal eventives and Eventives and Personal statives. Shaded areas denote 95% CrIs.

Deverbal vs. Nondeverbal eventives	0.01, [-0.05, 0.07]
Eventives vs. Personal statives	0.09, [0.03, 0.15]

Table 1: Posterior estimate and 95% CrIs for Event likelihood, C-command, and their interaction. Evidence for an effect is found if the 95% CrI excludes 0.

References: [1] Larson, R.K. (1998). *Proceedings of SALT VIII*, 145-168. [2] Winter, Y. & Zwarts, J. (2013). *Proceedings of IATL*, 1-14. [3] Rapp, I. (2015). *Journal of Semantics 32*, 477-523. [4] Forster, K.I., C. Guerrera & L. Elliot (2009). *Behavior Research Methods 41*(1), 163-171.